

Paloma Zarazua

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EDUCATION

M.S. Biology, California State Polytechnic University, Pomona Aug 2023- Dec 2025

Thesis: Investigating the Effects of Novel Metal Alloys on Neutrophil-Like Cell (NLCs) Responses

Advisor: Dr. Alas

B.S. Biology, University of La Verne, La Verne Aug 2019- Dec 2022

RESEARCH EXPERIENCE

Research Assistant, California State Polytechnic University, Pomona Nov 2025 - Present

Dr. Mercer's Laboratory & Dr. Lima's Laboratory, Biological Sciences Department

- Perform venipuncture on human research participants under IRB-approved protocols.
- Isolate primary neutrophils and monocytes from peripheral blood using density gradient centrifugation for downstream host-pathogen interaction studies involving *Toxoplasma gondii* and *Trichomonas vaginalis*.
- Conduct cell counts, viability assessments, and cryopreserve monocyte populations for downstream assays.
- Maintain biosafety compliance and human subjects research documentation.
- Coordinate sample processing workflows to preserve cellular integrity for experimental use.

Master's Research, California State Polytechnic University, Pomona Aug 2023 – Dec 2025

PI: Dr. Alas, Biological Sciences Department

- Designed and conducted experiments investigating innate immune cell responses to experimental metal alloys.
- Differentiated HL-60 cells into neutrophil-like, macrophage-like, and osteoclast-like phenotypes to model immune-biomaterial interactions.
- Developed and optimized cytokine assays and flow cytometry protocols.
- Performed statistical analysis in R (tidyverse, dplyr) and analyzed high-dimensional flow cytometry data using FlowJo.
- Interpreted inflammatory signaling patterns relevant to osteolytic and tissue remodeling mechanisms.

INDEPENDENT RESEARCH / COMPUTATIONAL PROJECTS

Independent Research Project- TCGA Lung Adenocarcinoma (LUAD) Analysis 2026

- Processed RNA-seq gene expression data from the Cancer Genome Atlas (TCGA-LUAD cohort) using R.
- Extracted and analyzed immune-related gene expression (CD8A, CD3D) across ~600 tumor samples.
- Performed data cleaning, transformation (log2 normalization), and quality control.
- Generated visualizations including boxplots and scatter plots using ggplot2
- Investigated T-cell infiltration signatures in lung adenocarcinoma tumors.

RESEARCH MENTORING

- Mentored 3 undergraduate and 2 graduate students in cell culture, sterile technique, microscopy, and flow cytometry.
- Trained students in HL-60 differentiation protocols and laboratory safety procedures.

TEACHING EXPERIENCE

Teaching Associate, Cal Poly Pomona

Aug 2023 - Dec 2025

- Delivered laboratory instruction and mini lectures for undergraduate biology courses (BIO 1110L, BIO 1150L, BIO 1220L, BIO 2350L).
- Instructed students in vertebrate dissections across multiple species.
- Trained students in molecular biology techniques including PCR, gel electrophoresis, and human cheek cell DNA extraction.
- Guided students in microscopy, specimen preparation, and morphological analysis.
- Taught statistical analysis concepts, including ANOVA using Microsoft Excel.
- Developed, administered, and graded weekly laboratory assessments.
- Supervised and mentored approximately 48 students per semester.
- Guided students in experimental design, data collection, and physiological data analysis (BioPac systems).

PRESENTATIONS

Cal Poly Pomona College of Science Research Symposium –May 2025

Neutrophil-Like Cells Response to Experimental Metal Alloys

CERTIFICATIONS & TRAINING

- **Certified Phlebotomy Technician I** (CDPH, active- 2027)
- **CDC Laboratory Training Modules** (PPE, Bloodborne Pathogens, Molecular Biology)
- **CITI Program:** Human Subjects Research, Animal Research (Mice), Laboratory Safety, Importance of Peer Review and Data Validation in Research, GDPR and Human Subject Research
- **NIH:** Ethical and Regulatory Aspects of Clinical Research Certificate
- **KGI Pre-Licensed Clinical Care and Patient Experience**
- **COPE Health Scholars Program Certificate**

TECHNICAL SKILLS

Cell & Immunology Techniques:

Primary neutrophil and monocyte isolation; density gradient centrifugation; HL-60 culture and differentiation; macrophage-like and osteoclast-differentiation, cell counting and viability assays; cryopreservation; sterile technique

Molecular Biology Techniques:

PCR; agarose gel electrophoresis; DNA extraction (human cheek cells); Western blotting (SDS-PAGE, membrane transfer, antibody detection)

Instrumentation & Analysis:

Flow cytometry (FlowJo); light microscopy; BioPac physiological data systems

Data & Statistical Analysis:

R(tidyverse, dplyr); one-way ANOVA in Microsoft Excel; basic data visualization and statistical interpretation

Regulatory & Clinical Research:

IRB-approved human subjects research; venipuncture; biosafety compliance; CITI certified

LANGUAGES

Spanish (fluent)

English (fluent)

ADDITIONAL EXPERIENCE

Phlebotomist**Nov 2025- Present**

California State Polytechnic University, Pomona

- Perform venipuncture on human research participants under IRB-approved protocols.

Patient Representative**Sept 2021- Oct 2022**

COPE Health Solutions- University of Texas Rio Grande

- Supported patient services and scheduling.

Health Scholar**Mar 2021- May 2022**

COPE Health Scholars- Emanate Health

- Assisted clinical staff and laboratory operations.